references 1 and 2, the deflector causes the free-stream flow to be deflected upward in the region ahead of the exhaust, whereas vortices formed near the flow deflector tips provide additional upward deflection of the free-stream flow at the outboard regions of the exhaust exit.

Flow deflector 2 (fig. 3(b)) differs from flow deflector 1 in that it was moved back on the aluminum supports 2.54 cm (1.0 in.) to place the leading edge of the deflector tips at an angle of 0° with respect to the model reference axis. The swirler was installed in the exhaust duct for this installation Figures 15 and 16 indicate no significant change in the impingement when compared with deflector 1 (fig. 12), but they do indicate a larger stalled or separated flow in the wake region aft of the flow deflector such as that noted for the spoiler (fig. 11). Figures 17 to 26 present the results of a more detailed investigation of flow deflector 2 without the swirler installed in the exhaust duct. The ratio of effective windstream velocity to exhaust velocity, angle of attack, and sideslip angle were varied as well as the position of the exterior source of bubbles. Figure 17 indicates that as the angle of attack is increased negatively, the impingement increases. However, figures 18, 19, and 20 which present simultaneous side and top views show that the exhaust flow is dispersing laterally well clear of the external exhaust system. Varying V_{eff} (figs. 21, 22, and 23) causes slight change in the angle of the exhaust flow patterns and on the dispersion of the flow. Photographs of sideslip conditions are presented for sideslip angles of 5° and -5° at $V_{eff} = 0.7$ (figs. 24, 25, and 26). Although larger angles of 10° and -10° were tested, the photographs disclosed no greater effects than those shown for 5° and -5°. In summary, varying the ratio of effective windstream velocity to exhaust velocity, angle of attack, and sideslip angle did not diminish the effectiveness of the flow deflector.

Flow deflector 2 was modified by adding clay to the leading edge of the center section of the flow deflector and was designated flow deflector 3. (See fig. 3(c).) The purpose of this modification was to reduce stalled airflow just aft of the center section of the flow deflector. Figures 27 and 28 indicate that there were no significant reductions of the stalled or separated flow region aft of the flow deflector in the wake. However, figure 27(d) is of special interest in that it shows the tip vortex core generated by the flow deflector tip.

Flow deflector 4 (fig. 3(d)) had a constant chord of 25.4 cm (10 in.) and a span of 76.2 cm (30 in.) which was cut into three equal span sections. These sections were joined together so that the outboard sections canted downward 15° to cause the tip vortices shed from the tips of the flow deflector to flow past the exhaust system nearer the level of the exhaust exit. Figures 29 and 30 show no significant advantage of this configuration. In fact, figure 29(b) indicates that there is a slight impingement; this impingement indicates that the placement of the tip vortex might be too low. The stalled or separated flow in the wake region aft of the center section still exists as shown in figure 29(c).